What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Nitrates

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0:00 / 16:50 1x

Learning Objectives (4)

After completing this brick, you will be able to:

1. Explain the mechanism of action of nitrates in the context of treating angina pectoris.
2. Categorize nitrates based on its route of administration, onset, and duration of action for acute vs chronic management of angina pectoris.
3. Explain the adverse effects of nitrate use.
4. Explain the c strategies to mitigate it.

CASE CONNECTION

A 65-year-old male, JD, was recently admitted to your medicine team with concern for unstable angina, and need for stress test. Initially, he presented to the ED with chest pain despite taking his isosorbide mononitrate this morning, about 4 hours before arriving to the ED. He has a history of hypertension, hyperlipidemia, and a myocardial infarction 5 years ago. He is retired, lives with his wife, and walks daily for exercise. His angina is usually controlled with his isosorbide mononitrate, which he takes daily.

As you come on service with the team for the evening and evaluate JD you note that it has been about 14 hours since his initial presentation to the ED, he has a transdermal nitroglycerin patch on his left chest, and he received sublingual nitroglycerin (0.4 mg x 3 doses) in the ED. His chest pain was well controlled, 2/10, throughout his ED stay and after being admitted. But, the nurse reports that JD has endorsed increasing chest pain, now 9/10, over the last 30-60 minutes.

Could JD's chest pain be medication related? Could the nitrate medications be losing efficacy? Let's keep JD in mind as we go through this brick and we will look to find an explanation for why his chest pain has suddenly increased.

GO TO CONCLUSION ↓

Before taking a look at the nitrate medications, let's take a brief look at chest pain (i.e., angina) to better understand why nitrates are such a common medication.

What is Angina?

Angina pectoris (i.e., angina) is defined as chest pain that is caused by myocardial ischemia (i.e, lack of oxygen leading to cardiac cell death). Atherosclerosis (i.e., plaque buildup) reduces the effective diameter of blood vessels, effectively reduces blood flow to the myocardium, and, consequently, oxygen delivery to the heart becomes impaired. As the plaques become unstable, grow, or other stressors are introduced it can lead to ischemia, which may cause chest pain. Angina is a type of ischemic heart disease (IHD), which also includes myocardial infarction. See Figure 1 below. There are three types of angina: stable, unstable, and vasospastic. As we discuss angina in the context of nitrate treatment, we will refer to unstable angina as "acute" and stable angina as "chronic." This is an important distinction for medication choice as it will impact which route of administration you would choose!

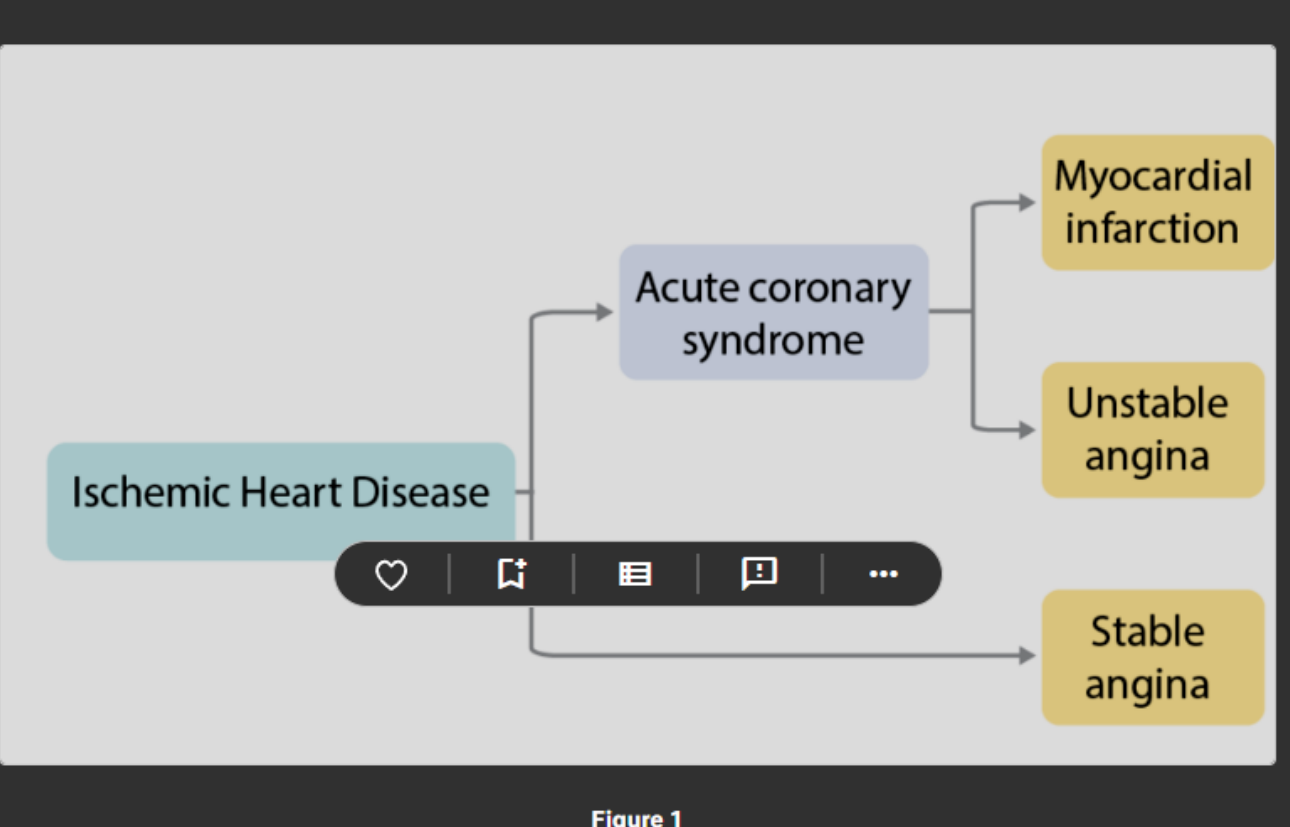


Figure 1

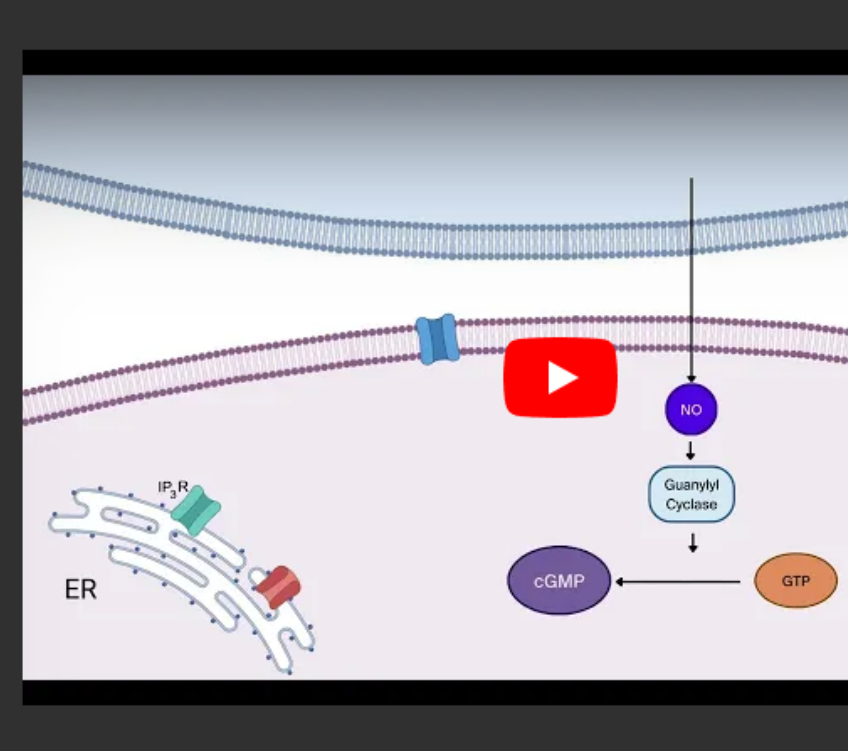
You will have the opportunity to discuss IHD and the types of angina in detail in the Cardiovascular Course, however, in this brick we will lay the foundation by focusing on how nitrates treat angina.

What causes angina?

How do nitrates treat angina? (LO#1)

Nitrates are clinically indicated for angina, both acute and chronic, and they are effective at reduc

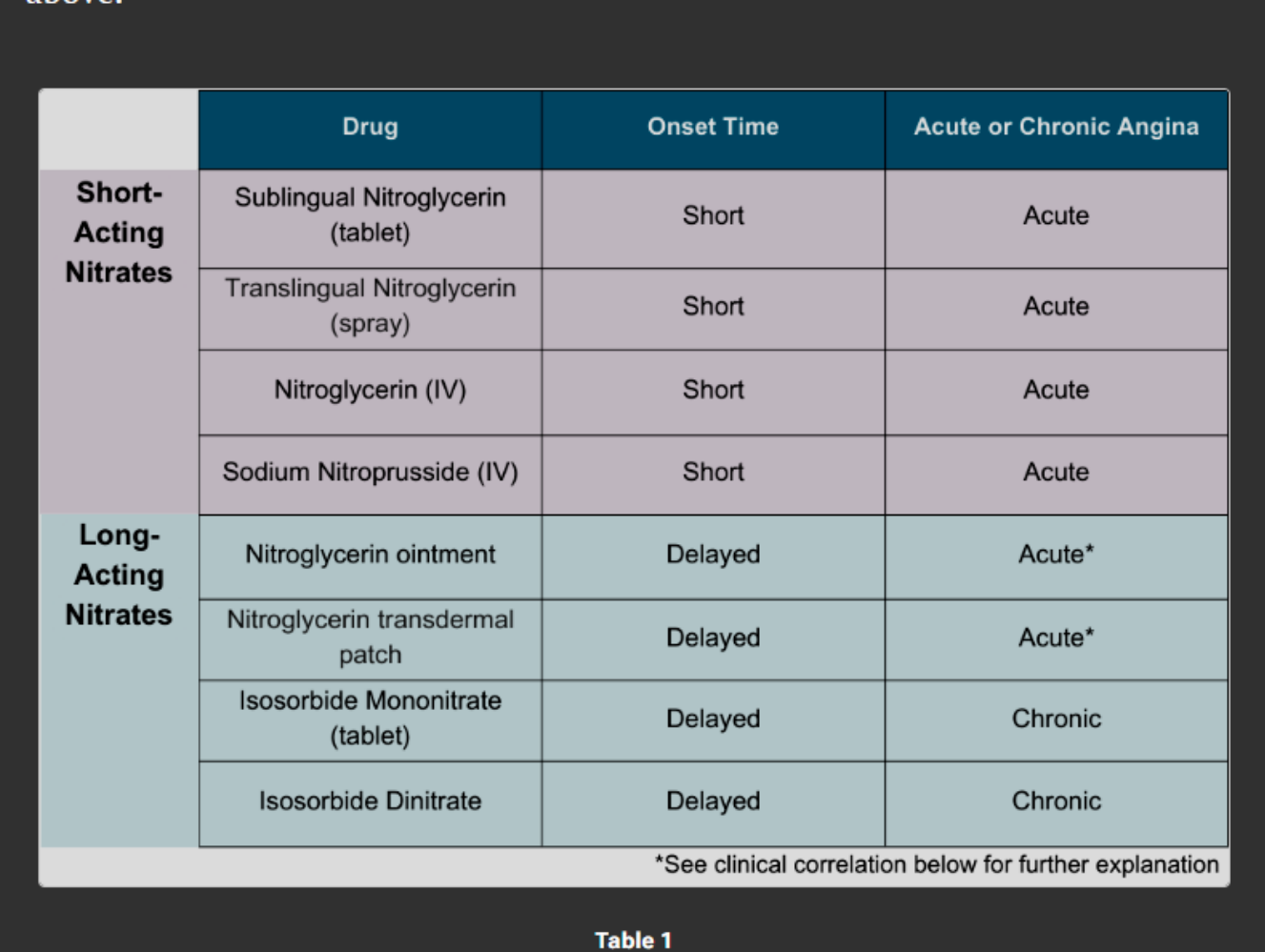
In the body nitrates are converted to nitric oxide (NO) which initiates the pathway discussed in the video below.



This relaxation, i.e., dilating, of the large veins that return to the heart **decreases contraction force**, **increases blood flow**, and **reduces myocardial oxygen** use, thereby reducing chest pain. In addition, nitrates cause some direct coronary artery vasodilation that increases oxygen supply to the heart, also relieving the chest pain.

How do we categorize nitrates? (LO#2)

Nitrates are available ration, have varying onset times (short vs delayed), and, based on their route of administration, they vary in whether they are used to treat acute vs chronic angina. Table 1 below categorizes and summarizes nitrates based on the parameters mentioned above.



	Drug	Onset Time	Acute or Chronic Angina
Short-Acting Nitrates	Sublingual Nitroglycerin (tablet)	Short	Acute
	Translingual Nitroglycerin (spray)	Short	Acute
	Nitroglycerin (IV)	Short	Acute
	Sodium Nitroprusside (IV)	Short	Acute
Long-Acting Nitrates	Nitroglycerin ointment	Delayed	Acute*
	Nitroglycerin transdermal patch	Delayed	Acute*
	Isosorbide Mononitrate (tablet)	Delayed	Chronic
	Isosorbide Dinitrate	Delayed	Chronic

*See clinical correlation below for further explanation

Table 1

Recall from week 1 of the course when we discussed absorption of medications. The route by which the medication is delivered to the patient can greatly alter the absorption of that medication, and then subsequently the effects of the drug are realized in different ways. For instance, sublingual nitroglycerin tablets are the most d under the tongue, they avoid first-pass metabolism, are rapidly absorbed, and can quickly relieve angina. On the other hand, isosorbide mononitrate is a delayed release tablet. After being swallowed by the patient, it is slowly absorbed, undergoes first pass metabolism, and, eventually, causes the same vasodilation effects. However, it is only used for chronic angina.

CLINICAL CORRELATION

Both Nitroglycerin ointment and the transdermal patch have a delayed onset of action. This, as mentioned in the paragraph above, is related to the transdermal absorption that they have, which delays their onset. This makes these medications particularly useful for the "late" acute angina. Let's look at a very common example you will encounter in the emergency department.

A 48 y.o. female experiences severe chest pain and calls 9-1-1. As EMS transports her to the nearest ED they give her sublingual nitroglycerin. As they arrive to the ED her chest pain is lessening, but still present. She get's another dose of sublingual nitroglycerin as the physician evaluates her. Again, it helps with the chest pain, but it comes back. Instead of continuing to re-dose sublingual nitroglycerin the patient could have nitroglycerin ointment, or a transdermal patch, applied to her chest. The delayed onset will provide a continuous effect as the medication is more slowly absorbed.

Why is sublingual nitroglycerin preferred for acute angina?

What are the adverse effects of nitrates? (LO#3)

The most common adverse effect associated with nitrates is *headache*. More than 60% of patients who receive nitroglycerin report headache. This number is significantly less in patients taking the long-acting isosorbide mononitrate. Headache likely occurs because of the rapid vasodilation that occurs with nitroglycerin. Also, because of the vasodilation, the drug can cause postural, or orthostatic, hypotension that may result in *dizziness and fainting*.

Another common adverse effect of nitrates is *hypotension*. In some instances, nitrates (i.e., IV nitroglycerin or sodium nitroprusside) are used to acutely lower blood pressure, but, as mentioned in the paragraph above, the significant vasodilation that occurs can cause hypotension. Any patient that is receiving nitrates should have their blood pressure monitored regularly.

What is the most common adverse effect of nitrates?

What is nitrate tolerance? (LO#4)

Frequently repeated or continuous exposure to high doses of nitrates leads to tolerance (i.e., tachyphylaxis), that is, marked attenuation in the magnitude of most of their pharmacological effects.

The magnitude of tolerance is a function of dosage and frequency of use. Tolerance may result from a reduced capacity of the vascular smooth muscle to convert nitroglycerin to NO, true vascular tolerance, or to the activation of mechanisms extraneous to the vessel wall.

Nitrate therapy should be designed to prevent tolerance. Nitrates **require a 10-12 hour nitrate-free interval** to decrease the likelihood of tolerance. This allows the return of efficacy. Some suggestions for reducing nitrate tolerance are listed below.

- Patch: Wear on for 12-14 hours, off for 10-12 hours. Rotate patch sites
- Ointment: Dosed twice daily, 6 hours apart with a 10-12 hour off period
- Isosorbide monor art

Why is a nitrate-free interval necessary in nitrate therapy?

CASE CONNECTION BACK TO INTRODUCTION ↑

Let's return back to JD! Based on what we have discussed in this brick, is it possible that JD is experiencing nitrate tolerance?

Remember that after 12-14 hours of continuous nitrate use patients can develop nitrate tolerance. This may result from a reduced capacity of the vascular smooth muscle to convert nitroglycerin to NO, true vascular tolerance, or to the activation of mechanisms extraneous to the vessel wall.

JD took his isosorbide mononitrate as prescribed in the morning, followed by sublingual nitroglycerin in the ED, and he has had a transdermal nitroglycerin patch on since that time. He has been receiving continuous nitrates for over 14 hours, and the return of his chest pain indicates that he is experiencing nitrate tolerance. In order t eds a 'nitrate free' period of 10-12 hours. In the cardiovascular organ system course you will discuss which other therapies would help relieve JD's chest pain, but it could include percutaneous coronary intervention in the cath lab, or potentially other medication therapy like beta-blockers or calcium channel blockers.

Summary

- Angina pectoris is chest pain caused by myocardial ischemia, often due to atherosclerosis.
- Nitrates treat angina by converting to nitric oxide in the body, which relaxes vascular smooth muscles, reducing myocardial oxygen use and relieving chest pain.
- Nitrates are available in various forms and routes of administration, each with different onset times and uses for acute or chronic angina.
- The route of administration can greatly alter the absorption and effects of the medication. For example, sublingual nitroglycerin tablets are rapidly absorbed and used for acute angina, while isosorbide mononitrate is slowly absorbed and used for chronic angina.
- Common adverse effects of nitrates include headache and hypotension due to rapid vasodilation.
- Continuous exposure to high doses of nitrates can lead to tolerance, reducing their efficacy. To prevent this, a 10-12 hour nitrate-free interval is needed.

Review Questio

1. Why does headache commonly occur with sublingual nitroglycerin?

A. ○ Delayed vasoconstriction
B. ○ Delayed vasodilation
C. ○ Rapid vasoconstriction
D. ● Rapid vasodilation

Hide Explanation

The correct answer is rapid vasodilation (D). As sublingual nitroglyucerin is rapidly absorbed and converted to NO it's downstream effects cause smooth muscle relaxation/ vasodilation. This rapid vasodilation can frequently cause headache. Sublingual nitroglycerin causes vasodilation, NOT vasoconstriction, this makes delayed vasoconstriction (A) and rapid vasoconstriction (C) incorrect answer choices, as explained in the mechanism above. And, as mentioned, above, the effects of sublingual nitroglycerin are rapid, making delayed vasodilation (B) an incorrect answer choice.

2. Which of the following is a strategy to mitigate nitrate tolerance?

A. ○ Increase the dosage
B. ● 10-12 hour nitrate-free interval
C. ○ Switch to a different class of medication
D. ○ Decrease the dosage
E. ○ Use nitrates continuously

Hide Explanation

The correct answer is to have a 10-12 hour nitrate-free interval (B). This allows the return of efficacy and decreases the likelihood of tolerance. Increasing the dosage (A) can lead to more severe side effects and does not necessarily prevent tolerance. Switching to a different class of medication (C) is not always possible or desirable, as nitrates are often very effective for treating angina. Decreasing the dosage (D) can lead to inadequate treatment of angina symptoms. Using nitrates continuously (E) can actually increase the likelihood of developing tolerance.

3. What is the primary mechanism of action of nitrates in treating angina pectoris?

A. ○ Increasing heart rate
B. ● Vasodilation through nitric oxide production
C. ○ Decreasing blood pressure
D. ○ Inhibiting platelet aggregation
E. ○ Blocking

Hide Explanation

The correct answer is vasodilation through nitric oxide (NO) production (B). Nitrates are converted to nitric oxide in the body, which then relaxes vascular smooth muscles, reducing myocardial oxygen use and relieving chest pain. Increasing heart rate (A) and decreasing blood pressure (C) are effects of nitrates, but not the primary mechanism of action. Inhibiting platelet aggregation (D) and blocking calcium channels (E) are not mechanisms of action of nitrates.

4. Which of the following nitrate medications undergoes first-pass metabolism, and is indicated for chronic angina?

A. ● Isosorbide mononitrate
B. ○ Nitroglycerin
C. ○ Sodium nitroprusside
D. ○ Sublingual nitroglycerin
E. ○ Transdermal nitroglycerin

Hide Explanation

The correct answer is medication with a delayed onset. Since it is oral it is subject to first-pass metabolism and is indicated for chronic angina. Nitroglycerin (B) and sodium nitroprusside (C) are IV medications that avoid first-pass metabolism and are indicated for acute angina. Sublingual nitroglycerin (D) and transdermal nitroglycerin (E) both avoid first-pass metabolism and are indicated for acute angina.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain the mechanism of action of nitrates in the context of treating angina pectoris.~~
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Objective Confidence: 4/4